

EXCEL PROJECT

YouTube Trending Video Performance & Engagement Analysis

Uncovering what drives content virality, audience reach, and engagement across global markets

123.71 B Total Views	India Top Market	Music Top Category	6 PM Peak Upload Hour
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1. PROJECT OVERVIEW

This project analyzes YouTube trending video data to uncover actionable insights across content performance, audience engagement, upload timing, and geographic reach — equipping content creators and digital teams with evidence-based strategies to drive visibility and growth.

Challenge	Content creators cannot reliably identify which variables — category, timing, geography — consistently drive trending status and sustained engagement.
Solution	Structured EDA in Excel quantifying the relationship between content attributes and performance KPIs across categories, countries, and time.
Value	Actionable benchmarks enabling teams to prioritize category investment, optimize scheduling, and replicate high-performing content patterns.

2. OBJECTIVES

- Analyze views by category to identify highest-ROI content types.
- Compare country-wise performance to surface priority geographic markets.
- Determine the optimal upload hour for maximum engagement velocity.
- Quantify the performance gap between trending and non-trending content.
- Evaluate likes and comments by category to distinguish passive vs. active engagement.
- Track monthly view trends to detect seasonality and inform launch planning.

3. DATASET ATTRIBUTES

Video ID / Title	Unique identifier for deduplication; title for qualitative pattern identification.
Category	Primary segmentation variable — Music, Vlogs, Sports, and more.
Publish Date	Decomposed into Month, Day Name, and Upload Hour for temporal analysis.
Views / Likes / Comments	Core engagement metrics — foundation for all KPI calculations.
Country	Geographic dimension for market-level audience analysis.
Trending Status	Binary flag (Trending / Non-Trending) — key comparative variable.
Engagement Rate	Pre-calculated metric validated against raw like and view counts.
Video Duration	Transformed into Short / Medium / Long buckets for format analysis.

4. DATA CLEANING

- Missing value audit: Identified and resolved null records across all key metric columns.
- Deduplication: Removed duplicate records using Video ID as the unique key.
- Type correction: Enforced correct types on date, integer, and decimal fields.
- Standardization: Normalized category and country values to eliminate pivot inconsistencies.
- Engagement rate validation: Cross-validated rates against raw likes/views for formula integrity.

Tools Used

Microsoft Excel | Pivot Tables | Pivot Charts | Excel Formulas (SUMIF, TEXT, IFERROR, VLOOKUP)

5. DATA TRANSFORMATION

Created New Calculated Columns

Month Name	Seasonal trend analysis and content launch planning.
Day Name	Day-of-week publishing pattern assessment.
Upload Hour	Peak engagement window identification.
Duration Minutes	Consistent cross-video format comparison.
Video Type (S/M/L)	Short / Medium / Long bucketing for audience preference analysis.
Calculated Engagement	Independently verified engagement rate for analytical reliability.

Developed KPIs

123.71 B Total Views	5032.6 M Total Likes	748.4 M Comments	3521 Trending Views count	7.75% Avg Engagement rate
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6. EXPLORATORY DATA ANALYSIS (EDA) & VISUALIZATION

Analysis Performed

- Category-wise performance — identifying top-performing content types by total views.
- Country-wise performance — ranking geographic markets by audience volume.
- Upload time analysis — mapping engagement rate against hour of publication.
- Trending vs. Non-Trending — quantifying the virality performance premium.
- Engagement analysis — comparing likes and comments across content categories.
- Monthly trend analysis — detecting seasonal peaks and troughs across the year.

Dashboard Features Included

KPI Cards | Pivot Tables | Pivot Charts | Slicers | Interactive Filters | Dynamic Dashboard

Visualizations Created

Column Chart	Views by Category — ranking content types by total audience reach.
Bar Chart	Country-wise Views — identifying priority geographic markets.
Line Chart (×2)	Monthly View Trends & Best Upload Hour Analysis.
Doughnut Chart	Trending vs. Non-Trending Videos — visualizing the virality gap.
Clustered Column	Likes and Comments by Category — passive vs. active engagement.

7. KEY INSIGHTS

Views by Category	Music generated 16.5B views — highest of any category — driven by universal appeal, repeat-view behavior, and strong algorithmic promotion.
Country-wise Views	India contributed 21.8B views, the largest single national audience. Creators without an India strategy are leaving significant reach unrealized.
Best Upload Hour	Videos published at 6 PM achieved the highest average engagement rate, aligning with peak post-work browsing and maximizing early engagement velocity.
Trending vs Non-Trending	Trending videos captured 86.9B views vs non-trending — a compounding effect where trending status amplifies reach far beyond organic performance.
Likes & Comments	Vlogs led in likes (685.7M) — strong community connection. Music (98.8M) and Sports (98.7M) drove highest comments, signaling active audience discussion.
Monthly Trend	May peaked at 18.9B views; July dipped to 16.3B — a 13.5% seasonal variance that enables predictable content launch planning.

8. BUSINESS RECOMMENDATIONS

#	Recommendation	Business Impact
1	Anchor strategy in Music	Music's 16.5B view lead makes it the highest-ROI category — allocate 40%+ of output to Music or Music-adjacent formats.
2	Develop India-first strategy	At 21.8B views, India is an outsized opportunity. Localized content and India-specific cadences unlock significant incremental reach.
3	Standardize 6 PM uploads	Zero-cost, immediately actionable. Implement as default publishing policy across all channels.
4	Engineer for trending eligibility	Analyze shared attributes of trending videos to build a trending-readiness checklist applied before every upload.
5	Use Vlogs for subscriber loyalty	Vlogs generate the highest like volume, signaling emotional connection — ideal for community growth and retention.
6	Music & Sports for community	High comment volumes fuel social amplification and the discussion signals that support algorithmic ranking.
7	Align launches with peak months	Time highest-investment releases to May. Use July off-peak periods for A/B testing new content formats.
8	Monthly KPI review cadence	Track trending rate, engagement rate, and country performance to detect audience behavior shifts early.

9. CONCLUSION

This project analyzed YouTube trending video data to identify key performance patterns. The findings showed that content category, country, publishing time, and trending status significantly impact viewership. Music emerged as the most popular category, while India contributed the highest volume of trending content. Videos published during peak evening hours generally achieved better engagement. The dashboard transformed raw data into meaningful insights through effective visualizations. Overall, this project demonstrates how data-driven decisions can improve content strategy and audience reach.